Jian Feng

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Skills

Machine Learning: PyTorch, Hugging Face Transformers, LoRA, scikit-learn, XGBoost
Methods: model fine-tuning, ensemble learning, statistical analysis, evaluation, error analysis
Programming: Python, Java, C/C++, SQL, TypeScript
Systems: Azure, GCP, Kubernetes, Docker, Git, Linux, REST, WebSockets

Experience

Manulife Financial Corporation Toronto, ON, Canada
Software Engineer *May 2026 – Aug 2026*

- Built and refined reusable ServiceNow workflows for **5K+ developers**, automating **1K+ daily operations**; integrated live AKS and certificate checks into a **Foundry LLM agent** to flag GitOps deployment mismatches.
- Replaced plaintext GitHub credentials with **GitHub App token exchange**, adding team-scoped validation, least-privilege permissions, and audit trails.

University of Waterloo Waterloo, ON
Research Assistant *Jan 2026 – Apr 2026*

- Fine-tuned four specialist DeBERTa safety models with LoRA and co-designed a learned aggregator; neutral-context training reduced medical/education false positives, with **sub-50ms inference on a T4**.
- Compared learned fusion with fixed-threshold baselines (**80% F1, 95%+ end-to-end accuracy on held-out data**) and red-teamed against Llama-3-8B on **5K prompts**.

Treasury Board Secretariat Toronto, ON, Canada
Software Developer *Sep 2025 – Dec 2025*

- Automated duplicate/invalid job-title cleanup in a **5K-firm dataset**, reducing manual processing time **85%**; tuned SQL to cut report load time **30%**.
- Migrated **30 Selenium tests across five suites** to Katalon on Azure DevOps CI/CD; executive demo drove adoption by **two QA teams**.

Waterloo Aerial Robotics Group Waterloo, ON
Autonomy Member *Jan 2025 – Apr 2025*

- Rebuilt telemetry for the UAV ground-station console (React/TypeScript, Python WebSocket, ROS), restoring **5 dashboard panels** and adding session record/replay for **4.2 hours** of flight logs within a **64 MiB** memory bound.

Lansi Institute of Artificial Intelligence in Medicine Remote
Data Analyst *May 2024 – Dec 2024*

- Parallelized preprocessing of **100GB of multi-omics data** with per-sample Python multiprocessing, cutting runtime **89%, from days to hours** for an international T2D research collaboration.

Sanyang Technology Co., Ltd. Guangdong, China
Software Engineer *Jun 2024 – Nov 2024*

- Translated business-analyst requirements into implementation design and built a Java/Spring job-shop scheduling algorithm, APIs, and interactive Gantt renderer end-to-end.
- Demoed the workflow, secured team sign-off, and tested against production data; exposed reporting through Spring Boot/MySQL APIs for dashboards and downstream ETL.

Education

University of Waterloo Sep 2023 - Aug 2027
Bachelor's degree, Computer Science, Finance, and Statistics *Waterloo, ON, Canada*

Projects

Army of Safeguards | Python, PyTorch

- Built a text-safety system with four specialist DeBERTa detectors for toxicity, sensitive content, jailbreaks, and factuality.
- Implemented configurable threshold, weighted, and learned per-axis aggregation with benchmark and training pipelines.

GhostPilot | Python, Windows

- Built a multimodal AI pipeline in a Windows overlay, routing WASAPI audio through Azure Speech or local ASR and combining transcripts with captured screen context for LLM responses.
- Integrated OCR/vision prompting and hybrid RAG over a user-curated knowledge base to ground answers in on-screen content and reference material.

Achievements

- **Velocity Co-op Problem Award** (top cohort submission); Moonshot Hackathon Top 100 finalist Award; WorldQuant Brain Gold Level (research consultant candidate)
- As UX Research Hub Director, owned community growth and programming end-to-end: ran member research, secured a Google UXR guest speaker (**50 attendees**, **50 new members**), grew membership **32× (10⟶318)**, and managed event funding with the executive team.